Airbnb Analytics Manager

Interview Prep Pack

Case-study frameworks, SQL practice with model answers, experimentation, Python/data modeling, presentation and behavioral prep. Built from our prep conversations.

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Caveats: Airbnb process details (rounds, SQL round format, take-home prompt) come from material you pasted, not verified by me. Industry notes are general knowledge, not sourced statistics. I have not analysed the take-home CSV; Section 6 is an approach, not findings. Airbnb values beyond the two you listed are from memory; verify on Airbnb's careers site.

1. Interview structure & plan

Round	Focus	What they look for
Product analytics & case	Marketplace dynamics, metric definition, root-cause analysis	Structured framework across supply, demand, macro/seasonality. Sample: 'Bookings down 5% WoW. Investigate.'
Experimentation & stats	A/B, causal inference, variance reduction	Network effects, seasonality, quasi-experiments, interrupted time series, seasonal controls
Technical design & Python	Metric logic, data modeling, transformations	Live pandas, facts vs dimensions, tricky tracking logic
Past work presentation	Communication, technical depth, end-to-end impact	Project lifecycle, structuring ambiguity, defending methodology under probing
Core values & leadership	Behavioral, ownership, culture	'Champion the Mission', 'Be a Host'. PM conflict, mentoring analysts, extreme ownership (highly weighted, peer-led)
SQL (separate round pasted)	5 questions, 60 min, medium-hard, marketplace context	Reason about data model first; explain trade-offs and scaling. Topics: window functions, occupancy, superhost logic

7-day plan

Day	Focus	Reuse
1	Product case: booking-drop RCA (Section 3)	Hike DAU-drop case
2	Marketplace metrics + metric design (Section 2)	Hike metric hierarchy, eBay take-rate
3	Experimentation: interference, DiD, synthetic control, ITS (Section 5)	eBay switchback/holdout notes; Meta holdout sizing
4	SQL + Python drills (Sections 7-8)	SQL strong; Python is the gap
5	Build the Meta project deck and rehearse probes (Section 9)	Meta STAR story
6	Values & stories (Section 10)	Hike scenario categories, 90-day plan
7	Mock the two weakest rounds, fix gaps	-

Gaps to close

- Interrupted time series: not yet practised.
- Python live coding: needs timed practice.
- Airbnb domain: supply/demand liquidity, host behavior.
- Meta project: needs hard result numbers (budget scaled by X%, savings Y).

Questions to ask your contact who interviewed

Process & structure

1. Exact sequence of rounds and elapsed time from recruiter call to offer?
2. Length of each round and who interviewed (peer, manager, cross-functional)?
3. Was there a recruiter/HM screen first? What did it cover?
4. Single day or spread out? Which round surprised you?
5. Presentation round: time, panel size, prep time given, Q&A; split?

Per round

1. SQL: dialect, live vs doc, how many questions actually asked, scaling/trade-off follow-ups?
2. Product case: Airbnb-specific or generic? How structured did they expect you to be?
3. Experimentation: depth on network effects, ITS, quasi-experiments? Any math?
4. Python: pandas live coding or data-model/architecture? Whiteboard?
5. Past work: did they pick the project or you? How hard was the methodology probing?
6. Core values: which values were tested? Which behavioral questions came up?

Manager-specific & evaluation

1. How much people-management evaluation (team building, mentoring, hiring)?
2. Was the bar different for manager vs IC in technical rounds?
3. What separated strong from average answers in her view?
4. What did she wish she had prepared more? Which resources helped?
5. Any feedback she received afterwards? Level calibration conversation?

If she is under NDA, ask about process, depth and style, not exact questions.

2. Marketplace & travel metrics cheat sheet

Metric	Definition	Why it matters / watch-outs
Occupancy	Booked nights / (available + booked nights)	Ratio: rises if numerator up OR denominator down (hosts blocking calendar). Always inspect both.
Nights booked	Sum of confirmed nights stayed in window	Volume. Define by booking date vs check-in date; they tell different stories.
GBV / take rate	Total booking value; Airbnb fee share of it	Revenue levers: volume x price x take rate.
ADR	Average daily rate = revenue / booked nights	Price level. Mix shift (more cheap rooms) lowers ADR without any price change.
RevPAR-style	Revenue per available night = ADR x occupancy	Combines price and utilization; common hotel-industry KPI.
Lead time	Days between booking and check-in	Shortening lead time can shift when bookings appear and make WoW comparisons misleading.
Length of stay	Nights per booking	Longer stays raise nights per booking and occupancy; also lower turnover.
Search-to-book	Bookings / search sessions	Funnel conversion; split by market and supply tightness.
Liquidity	% of searches with enough matching available listings	Marketplace health: both sides must clear.
Host acceptance / response	Accepted requests / requests; replies within SLA	Supply-side friction; Superhost criteria.
Cancellation rate	Cancelled / confirmed, split host vs guest	Host-initiated cancellations hurt trust far more.
Repeat guest rate	Guests with another booking within N days	Retention; right-censoring for recent cohorts.
Host retention	Hosts still active after N months	Supply health; new vs churned hosts affect occupancy composition.

Marketplace concepts to name

- Two-sided: actions on one side change outcomes on the other (interference/SUTVA violations).
- Seasonality: weekly, monthly, holiday, school-term; compare YoY same-week, not just WoW.
- Mix shift vs rate change (Simpson's paradox): an aggregate can move purely from composition.
- Survivorship: low performers delist, so surviving hosts look better.
- Guardrails for any test: cancellations, reviews, support contacts, host/guest retention.

3. Product case frameworks

Opening line for any case: 'Before I dive in, I'd confirm the metric definition and whether this is real or a measurement artifact, then split by supply vs demand, and rank hypotheses by size and likelihood.'

Case A: 'Bookings are down 5% week-over-week. How do you investigate?'

Step	What to do / say
0. Clarify	Which metric (bookings, nights, GBV)? By booking date or check-in date? Which weeks, is the week complete? Global or localized? Any launches, outages, campaigns?
1. Validate data	Pipeline health, event-tracking changes, definition changes, late-arriving data (cancellations and refunds post later). Compare server-side vs client-side counts.
2. Seasonality & calendar	Compare YoY same week and prior-year WoW. Holiday shifts, day-of-week mix, school terms, weather, big events. 'A 5% dip is normal after a peak; unusual in a shoulder week.'
3. Decompose	Bookings = traffic x search-to-book x acceptance x (1 - cancellation). Demand: traffic by channel, new vs returning. Supply: active listings, availability, price, acceptance. Funnel: visit, search, view, request, confirm, stay.
4. Segment	Geo/market, platform, device, trip type (LOS, lead time, party size), room type, price band, guest tenure, acquisition channel. Look for one segment over-indexing on the drop.
5. Hypothesis tree	Demand-side (traffic, paid spend, SEO, competitor), supply-side (fewer listings, calendar blocking, higher prices, more host declines), friction (bug, payment failures, search ranking), external (macro, regulation, events). For each: the cut that would confirm and the result that would refute it.
6. Quantify	Contribution table: share of the 5% drop explained by each segment/funnel step. Separate mix vs rate effects.
7. Recommend	Fix now (bug, ranking, pricing nudges), monitor (watch metrics with thresholds), investigate further (what data is missing). Owner and expected recovery.
8. Prevent	Anomaly alerts on funnel steps by market; weekly health review; pre-agreed rollback triggers.

Case B: 'Occupancy is up YoY but there were no product changes. Why?' (mirrors take-home)

Step	What to do / say
0. Clarify	Occupancy = booked nights / (available or booked nights). It is a ratio: ask which side moved. Time window, markets, hosts included.
1. Mechanical: denominator	Hosts blocking low-demand dates, or delisting weak listings, shrinks the denominator and lifts occupancy with no extra demand. Check total available nights, listing counts, calendar blocking.
2. Mechanical: composition	Mix shift toward high-occupancy segments (room type, region, capacity) or new vs churned hosts. Run a mix vs within-segment decomposition (Section 7, A3).
3. Real demand	Travel rebound, market events, longer stays (more nights per booking), shorter lead times, pricing changes by hosts, seasonality shift.
4. Supply behavior	Hosts discounting, lowering minimum stays, becoming more professional (multi-listing hosts). Compare listings that were active in both periods (balanced panel) if listing/host IDs exist.
5. Test each hypothesis	State the cut and the refuting result: e.g. 'If the rise is within every segment, mix is ruled out; if available nights fell while booked nights were flat, it is a denominator story.'
6. Implication	Higher occupancy is not automatically good: if it came from calendar blocking, hosts earn the same with less availability and guests see fewer options. Pair with revenue per available night and availability.
7. Recommendation	Depends on cause: if supply shrinkage, nudge hosts to open calendar; if real demand, support pricing guidance; track occupancy alongside availability and ADR.

4. Travel case studies

Universal skeleton: **Clarify, define success metric + guardrails, hypothesis tree, data/cuts, diagnose or design, quantify, recommend, measure.** Show second-order effects and a valid counterfactual.

Case 1: Host-initiated cancellations up 20% in a top city

Step	What to do / say
Clarify	Rate or count? Host-initiated only? Time of cancellation relative to check-in? Which host segments? Since when?
Why it matters	Guest trust, rebooking cost, review score hits, support load, repeat rate. Quantify guest impact: rebooking rate, reviews after a host cancel, 90-day return.
Hypotheses	Event-driven demand spike (hosts cancel to rebook at higher price), calendar sync problems (double-booking), new hosts without experience, policy loophole (low penalty), regulatory or maintenance issues, pricing tool changes.
Diagnose	Cancellation reasons if captured; days-before-checkin distribution; price delta between cancelled and subsequent booking of same dates; host tenure, multi-listing, channel-manager use; clustering around events.
Policy options	Higher penalties, calendar-sync tooling, stricter rules for repeat offenders, guaranteed rebooking, ranking penalties.
Measure	Experiment at host or market level (interference risk: guests substitute across listings). Primary: host-cancel rate. Guardrails: supply (active listings, new host sign-ups), host satisfaction, bookings.

Case 2: Launch a smart-pricing suggestion tool for hosts

Step	What to do / say
Clarify objective	Host revenue, nights booked, occupancy, or marketplace conversion? Different north stars lead to different tools.
Success metrics	Primary: host revenue per available night. Secondary: adoption, occupancy, ADR. Guardrails: guest conversion, review score, cancellations, price dispersion, host trust.
Design problem	Hosts in the same market compete for the same guests: control hosts are affected by treated hosts' price changes (interference). Randomize at market or cluster level, or use switchback by market-time; accept less power.
Selection bias	Hosts who adopt suggestions differ (more engaged). Analyze intent-to-treat; use adoption only for descriptive heterogeneity.
Long-term	Price convergence across hosts, effect on guest price perception, take-rate revenue. Keep a long-run holdout market.
Decision rule	Pre-register MDE and kill thresholds before launch.

Case 3: City caps short-term rentals; supply falls 25%

Step	What to do / say
Clarify	Cap date, rule type, which listings affected, neighboring markets.
Counterfactual	No randomization. Synthetic control from comparable cities (weights fit on pre-period); DiD with matched cities if trends are parallel; check pre-period fit and placebo tests.
Outcomes	Bookings, nights, prices (ADR), occupancy on remaining listings, host exits vs conversions to long-term rentals, guest demand displacement.
Spillover	Demand moves to neighboring cities: those are not clean controls. Model displacement explicitly or exclude.
Second-order	Remaining hosts have higher occupancy and ADR (selection); guests face fewer options; trust and regulation relationship.
Deliver	Effect estimate with intervals, sensitivity to donor pool, and what it implies for other regulated markets.

Case 4: Launch Experiences to existing guests

Step	What to do / say
North star	Incremental bookings and guest retention, not Experiences views. Decide if Experiences is a standalone product or a retention driver for stays.
Metrics	Attach rate, repeat-stay rate for Experiences users, GBV per guest. Guardrails: stay conversion, support contacts, reviews.
Cannibalization	Does it shift time/spend away from stays or add to it? Compare matched users or randomized exposure.
Incremental vs shifted	Randomize exposure (holdout), measure 90-day total spend per guest, not just Experience bookings.
Supply side	Experience host onboarding, quality, cancellation rates; supply constraints cap the effect.

5. Experimentation & causal inference

Experiment design framework (use every time)

Step	What to do / say
1. Decision	What decision will this inform? Define ship/no-ship rule up front.
2. Unit of randomization	Guest, host, listing, market or time window. Ask first. In a two-sided marketplace, guest-level randomization of a supply-side feature contaminates control.
3. Metrics	One primary (OEC), secondary diagnostics, guardrails as kill switches. Set before launch; never redefine after seeing results.
4. Power	MDE is a business decision. Sample size scales with variance / effect^2. Revenue metrics are high variance: longer runs or CUPED.
5. Duration	Whole weeks (day-of-week), check novelty decay, avoid peak seasons, pre-commit horizon; no peeking.
6. Validity checks	Sample ratio mismatch, pre-period balance, logging health.
7. Analysis	Intent-to-treat; absolute lift with CI; segments pre-registered; multiple-testing correction.
8. Decision & learning	Ship / iterate / kill; log it in a test registry.

Situation	Approach	Main risk
Guest-facing UI, low supply interaction	Guest-level A/B	Low
Host tool or pricing, shared inventory	Host or listing cluster, or market-level	Fewer units, less power
Search ranking (both sides)	Geo cluster or switchback	Markets not comparable / carryover
Platform-wide policy	Market holdout + DiD	Few markets, revenue cost
Shared capacity/infrastructure	Switchback with washout	Carryover between periods

Quasi-experimental methods

Method	Use when	Key assumption / checks
Difference-in-differences	Treated and untreated groups, pre/post data	Parallel trends: plot pre-period; placebo tests; two-way fixed effects
Synthetic control	One treated unit (a city), many donors	Good pre-period fit; placebo in donors; long pre-period
Regression discontinuity	Hard eligibility threshold	No manipulation (density test); covariate smoothness; local estimate only
Interrupted time series	Single series, intervention at known date, no control group	No concurrent shocks; model seasonality; autocorrelation-robust errors; better with a comparison series
Propensity matching	Observational selection only	Cannot handle unobserved confounders; check balance (SMD < 0.1) and common support

Interrupted time series: how to answer

Model the outcome as a function of a pre-trend, a level shift at the intervention date, and a slope change after it, plus seasonality controls (month/week dummies or Fourier terms). Use Newey-West or ARIMA errors for autocorrelation. Weakness: any other event at the same date is confounded. Strengthen with a comparison series (comparative ITS) or placebo dates.

```
y_t = b0 + b1*t + b2*post_t + b3*(t - T0)*post_t + seasonality_t + e_t
b2 = immediate level change      b3 = change in trend after intervention
```

CUPED and common stats traps

- CUPED: adjust metric with its pre-experiment value (theta = cov(Y, X)/var(X)); same power with less data.
- Peeking and post-hoc metric redefinition are p-hacking; extend only by pre-registered sequential rules.

- Novelty effects: look for decaying treatment effect inside the window.
- Survivorship in long tests; cannibalization: measure platform-level, not only in-feature.
- Seasonal controls: year-over-year comparison, week-of-year fixed effects.

6. Take-home: occupancy increase analysis

Prompt (pasted): Advanced Analytics, host success. Occupancy trend rising, cause unknown, no major product changes. Occupancy = booked nights / available or booked nights. Each row = monthly metrics for a host (hosts may have multiple listings). 24 hours, ~5 slides for an executive, one PDF with code and slides, do not share the data.

Dataset columns (observed): room_type, person_capacity, space_type, region, country, total_booked_nights, total_available_or_booked_nights, month (format like 1/1/17). About 19K rows. No host or listing ID is visible, so a same-host balanced panel is not possible; state this as a limitation.

Analysis steps

Step	What to do / say
1. Data audit	Parse month; check range, nulls, zero denominators, booked > available, duplicates, rows per month (growth of host count).
2. Trend	Platform occupancy by month two ways: sum(booked)/sum(avail) (weighted, matches the definition) and average of host ratios. Explain differences.
3. Decompose numerator vs denominator	Plot total booked nights and total available nights separately. Is the rise from more booked nights or fewer available?
4. Seasonality	YoY comparison by calendar month; separate seasonal pattern from trend.
5. Segment cuts	Occupancy by room_type, space_type, person_capacity, region, country over time.
6. Mix vs within-segment	Decompose YoY change into mix and within-segment effects (Section 7, A3). Rank contributors.
7. Distribution shift	Host-level occupancy distribution: more hosts at 100%? fewer near-zero hosts? Growing share of low-denominator rows?
8. Conclusion	Classify the rise: compositional, denominator-driven, or genuine demand; quantify each share.

Slide outline (5 slides)

1. Headline: occupancy rose X pp; the answer in one sentence and what it means for hosts.
2. Trend chart with seasonality: what rose, when.
3. Where it came from: numerator vs denominator and mix vs within-segment waterfall.
4. Segment deep-dive: which room types/regions drive it; supporting evidence.
5. Recommendations and next steps: actions, metrics to monitor, data gaps and caveats.

Presentation tips: one message per slide, plain-language titles, no jargon without definition, state assumptions. Include code in the appendix of the PDF.

7. SQL practice with model answers

Approach every SQL question: state the grain of each table and the output, list edge cases (cancelled status, NULLs, zero denominators, overlaps, time zones), define metrics in words, write CTEs, then discuss scaling (partitioning by date, pre-aggregation, avoiding fan-out joins, approximate counts). Dialect below: Postgres-style.

Round 1 questions (as pasted)

Q1. Occupancy rate per city, past 90 days (Medium, 14 min)

Tables: bookings(listing_id, checkin_date, checkout_date, status), listings(listing_id, city, available_days_per_month). Occupancy = booked nights / available nights. Interviewer tests: nights from checkin/checkout, filter confirmed, division by zero. Also clip bookings that straddle the window edge.

```
WITH params AS (SELECT CURRENT_DATE - 90 AS ws, CURRENT_DATE AS we),
booked AS (
  SELECT b.listing_id,
         SUM(GREATEST(0, LEAST(b.checkout_date, p.we) - GREATEST(b.checkin_date, p.ws))) AS booked_nights
  FROM bookings b CROSS JOIN params p
  WHERE b.status = 'confirmed'
    AND b.checkin_date < p.we AND b.checkout_date > p.ws      -- overlaps the window
  GROUP BY b.listing_id)
SELECT l.city,
       SUM(COALESCE(k.booked_nights, 0))                        AS booked_nights,
       SUM(l.available_days_per_month * 3.0)                    AS available_nights,   -- 90 days = 3 months
       SUM(COALESCE(k.booked_nights, 0))
         / NULLIF(SUM(l.available_days_per_month * 3.0), 0)     AS occupancy_rate
FROM listings l LEFT JOIN booked k USING (listing_id)           -- keeps listings with no bookings
GROUP BY l.city;
```

- Assumption to state: whether available_days_per_month already excludes booked nights; the 3.0 multiplier is an approximation for 90 days.
- Ratio of sums per city (not average of listing ratios) to weight by capacity; mention both.
- Scaling: pre-aggregate by listing before the city join; partition bookings by checkin_date; expanding to a nightly calendar table is exact but larger.

Q2. Superhost eligibility (Medium, 16 min)

Criteria in last 12 months: >=10 completed stays, >=90% response rate, >=80% 5-star reviews, zero host-initiated cancellations. The pasted prompt was cut off after 'initiated by hos...'; the schema below is the generic Part B schema.

```
WITH stays AS (
  SELECT l.host_id, COUNT(*) AS completed_stays
  FROM bookings b JOIN listings l USING (listing_id)
  WHERE b.status = 'completed' AND b.checkout_date >= CURRENT_DATE - INTERVAL '12 months'
  GROUP BY l.host_id),
resp AS (
  SELECT host_id, AVG(CASE WHEN responded_at IS NOT NULL THEN 1.0 ELSE 0 END) AS response_rate
  FROM host_messages WHERE received_at >= CURRENT_DATE - INTERVAL '12 months'
  GROUP BY host_id),
rev AS (
  SELECT l.host_id, AVG(CASE WHEN r.rating = 5 THEN 1.0 ELSE 0 END) AS five_star_rate
  FROM reviews r JOIN bookings b USING (booking_id) JOIN listings l USING (listing_id)
  WHERE b.checkout_date >= CURRENT_DATE - INTERVAL '12 months'
  GROUP BY l.host_id),
hc AS (
  SELECT l.host_id, COUNT(*) AS host_cancels
  FROM bookings b JOIN listings l USING (listing_id)
  WHERE b.status = 'cancelled' AND b.cancelled_by = 'host'
    AND b.checkin_date >= CURRENT_DATE - INTERVAL '12 months'
  GROUP BY l.host_id)
SELECT s.host_id
FROM stays s JOIN resp USING (host_id) JOIN rev USING (host_id) LEFT JOIN hc USING (host_id)
WHERE s.completed_stays >= 10 AND resp.response_rate >= 0.90
  AND rev.five_star_rate >= 0.80 AND COALESCE(hc.host_cancels, 0) = 0;
```

- Pre-aggregate each metric to host grain before joining: joining raw reviews and messages would fan out and distort counts.

- Clarify: response rate usually within a time SLA; whether hosts with no reviews qualify (INNER JOIN excludes them); rating scale.

Part A: this dataset

Table eda(room_type, person_capacity, space_type, region, country, total_booked_nights, total_available_or_booked_nights, month). Parse month to a DATE first. Each row is a host-month.

A1. Monthly platform occupancy (easy)

```
SELECT month,
       SUM(total_booked_nights) * 1.0 / NULLIF(SUM(total_available_or_booked_nights), 0) AS occ_weighted,
       AVG(total_booked_nights * 1.0 / NULLIF(total_available_or_booked_nights, 0))       AS occ_avg_of_hosts
FROM eda GROUP BY month ORDER BY month;
```

Why they differ: the weighted version matches the stated definition (nights-weighted); the average treats a host with 3 available nights like one with 30. A growing share of tiny hosts can move the average without moving real occupancy.

A2. Segment occupancy, availability share, YoY change (medium)

```
WITH m AS (
  SELECT room_type, month, SUM(total_booked_nights) AS b, SUM(total_available_or_booked_nights) AS a
  FROM eda GROUP BY room_type, month),
t AS (SELECT month, SUM(a) AS total_a FROM m GROUP BY month),
x AS (
  SELECT m.room_type, m.month, m.b * 1.0 / NULLIF(m.a, 0) AS occ, m.a * 1.0 / t.total_a AS avail_share
  FROM m JOIN t USING (month))
SELECT room_type, month, occ, avail_share,
       LAG(occ, 12) OVER w AS occ_prior_year,
       (occ - LAG(occ, 12) OVER w) * 100 AS yoy_pp
FROM x
WINDOW w AS (PARTITION BY room_type ORDER BY month);
```

Caveat: LAG(12) assumes no missing months; otherwise self-join on month - INTERVAL '1 year'.

A3. Mix vs within-segment decomposition (medium-hard)

Occupancy = sum over segments of (share of available nights w) x (segment occupancy o). The YoY change splits exactly per segment: within = avg(w) x delta(o); mix = delta(w) x avg(o). The two sum to w1*o1 - w0*o0 for each segment, so contributions add up to the total change.

```
WITH tagged AS (
  SELECT *, CASE WHEN month BETWEEN :cur_start   AND :cur_end   THEN 'cur'
                 WHEN month BETWEEN :prior_start AND :prior_end THEN 'prior' END AS period
  FROM eda),
seg AS (
  SELECT room_type, region, period,
         SUM(total_booked_nights) AS b, SUM(total_available_or_booked_nights) AS a
  FROM tagged WHERE period IS NOT NULL GROUP BY room_type, region, period),
tot AS (SELECT period, SUM(a) AS total_a FROM seg GROUP BY period),
sh AS (
  SELECT s.*, s.b * 1.0 / NULLIF(s.a, 0) AS occ, s.a * 1.0 / t.total_a AS w
  FROM seg s JOIN tot t USING (period)),
pv AS (
  SELECT room_type, region,
    MAX(CASE WHEN period='prior' THEN occ END) AS o0, MAX(CASE WHEN period='cur' THEN occ END) AS o1,
    COALESCE(MAX(CASE WHEN period='prior' THEN w END), 0) AS w0,
    COALESCE(MAX(CASE WHEN period='cur'   THEN w END), 0) AS w1
  FROM sh GROUP BY room_type, region),
f AS (   -- segment missing in a period: carry the other period's occupancy so 'within' = 0
  SELECT room_type, region, w0, w1, COALESCE(o0, o1) AS o0, COALESCE(o1, o0) AS o1 FROM pv)
SELECT room_type, region,
       100 * (w0 + w1) / 2 * (o1 - o0) AS within_pp,
       100 * (w1 - w0) * (o0 + o1) / 2 AS mix_pp,
       100 * (w1 * o1 - w0 * o0)        AS total_pp
FROM f ORDER BY ABS(100 * (w1 * o1 - w0 * o0)) DESC;
```

Check: SUM(total_pp) equals the platform occupancy change in pp. Say this out loud: it shows the decomposition is exact, not approximate.

Part B: generic marketplace schema

```
listings(listing_id, host_id, city, created_at)
bookings(booking_id, listing_id, guest_id, checkin_date, checkout_date, status, cancelled_by)
```

```
reviews(booking_id, rating)
host_messages(host_id, inquiry_id, received_at, responded_at)
```

B1 is Q2 above.

B2. Average gap between bookings per listing; flag any gap > 30 days

```
WITH ordered AS (
  SELECT listing_id, checkin_date, checkout_date,
         LAG(checkout_date) OVER (PARTITION BY listing_id ORDER BY checkin_date) AS prev_checkout
  FROM bookings WHERE status = 'confirmed')
SELECT listing_id,
       AVG(checkin_date - prev_checkout)        AS avg_gap_days,
       MAX(checkin_date - prev_checkout) > 30   AS has_gap_over_30
FROM ordered
WHERE prev_checkout IS NOT NULL AND checkin_date >= prev_checkout   -- drop overlaps (data issue)
GROUP BY listing_id;
```

Say: overlapping confirmed bookings indicate a data problem; flag them separately. Using LAG on the prior row's checkout can miss a long earlier booking; a running MAX(checkout) is safer.

B3. 90-day repeat rate by first-booking city and checkout month

```
WITH conf AS (
  SELECT b.guest_id, b.checkin_date, b.checkout_date, l.city,
         ROW_NUMBER() OVER (PARTITION BY b.guest_id ORDER BY b.checkin_date) AS rn
  FROM bookings b JOIN listings l USING (listing_id)
  WHERE b.status IN ('confirmed', 'completed')),
fb AS (SELECT * FROM conf WHERE rn = 1),
rp AS (
  SELECT f.guest_id,
         MAX(CASE WHEN c.checkin_date > f.checkout_date
                  AND c.checkin_date <= f.checkout_date + 90 THEN 1 ELSE 0 END) AS repeated
  FROM fb f LEFT JOIN conf c ON c.guest_id = f.guest_id AND c.rn > 1
  GROUP BY f.guest_id)
SELECT f.city, DATE_TRUNC('month', f.checkout_date) AS cohort_month,
       COUNT(*) AS guests, AVG(rp.repeated) AS repeat_rate_90d
FROM fb f JOIN rp USING (guest_id)
WHERE f.checkout_date + 90 <= CURRENT_DATE     -- exclude right-censored cohorts
GROUP BY f.city, DATE_TRUNC('month', f.checkout_date);
```

Scaling talking points

- Partition large fact tables by date; filter early; pre-aggregate to the join grain.
- Avoid joining two many-side tables directly (fan-out).
- Window functions on 1B rows: ensure partition keys are balanced; consider incremental daily rollups.
- Approximate distinct counts (HLL) for guest/host uniques when exactness is not required.

8. Python & technical design

Pandas drills

Weighted occupancy and mix decomposition

```
import pandas as pd
df['month'] = pd.to_datetime(df['month'], format='%m/%d/%y')
m = df.groupby('month').agg(b=('total_booked_nights','sum'), a=('total_available_or_booked_nights','sum'))
m['occ'] = m['b'] / m['a']

seg = (df.assign(period=lambda d: d['month'].dt.year)
         .groupby(['period','room_type','region'])
         .agg(b=('total_booked_nights','sum'), a=('total_available_or_booked_nights','sum')).reset_index())
seg['w'] = seg['a'] / seg.groupby('period')['a'].transform('sum')
seg['occ'] = seg['b'] / seg['a']
p = seg.pivot_table(index=['room_type','region'], columns='period', values=['w','occ']).fillna(0)
```

Sessionization (30-min inactivity)

```
ev = ev.sort_values(['user_id','ts'])
gap = ev.groupby('user_id')['ts'].diff() > pd.Timedelta(minutes=30)
ev['session_id'] = gap.groupby(ev['user_id']).cumsum()
```

Funnel conversion

```
steps = ['search','view','request','book']
f = (ev[ev.event.isin(steps)].drop_duplicates(['user_id','event'])
       .pivot_table(index='user_id', columns='event', aggfunc='size', fill_value=0)[steps] > 0)
conv = f.sum() / f['search'].sum()
```

Cohort retention

```
first = df.groupby('user_id')['month'].transform('min')
df['age'] = (df['month'].dt.to_period('M') - first.dt.to_period('M')).apply(lambda x: x.n)
ret = df.groupby([first.rename('cohort'), 'age'])['user_id'].nunique().unstack(fill_value=0)
ret = ret.div(ret[0], axis=0)
```

Data modeling: facts vs dimensions

Table	Grain	Notes
fact_booking	One row per booking (state changes tracked)	Status history table or SCD for confirmed, cancelled_by, timestamps
fact_listing_night	One row per listing per night (calendar)	Available / blocked / booked, price; makes occupancy exact
fact_search_event	One row per search or impression	Funnel and liquidity metrics
dim_listing	One row per listing version (SCD type 2)	Room type, capacity, city; changes over time
dim_host / dim_guest	One row per host / guest	Tenure, segment
dim_date	One row per date	Holiday, season flags

Tricky tracking logic to name

- Idempotent, de-duplicated event ingestion; late-arriving data and backfills.
- Cancellations as state changes: avoid double counting; define bookings by booking date vs stay date.
- Time zones: listing local time vs UTC for check-in dates.
- Overlapping or invalid calendar blocks; host calendar sync issues.
- Metric definitions in one governed layer (glossary + SQL + owner).

9. Past work presentation: Meta incrementality project

Structure (15-20 min): stakes, question, design, results, decision, what you'd change. Lead with the punchline.

STAR

Step	What to do / say
Situation	Flipkart spends heavily on Meta performance marketing; ROI relied on last-touch attribution, which overstates channel credit. Need: true incremental ROI. (Add Cr/year spend if you can.)
Task	Own the measurement design: a causal framework granular enough for BU and category budget decisions.
Action	Designed holdout A/B with 8% holdout; sized via power analysis on customer-conversion for 2-week and 3-week windows. Weekly readouts; stat-sig on incremental customers in ~2-3 weeks, on incremental GMV ~1 week later. Split by BU (each owns spend), computed incremental customers, incremental GMV, ROI. Drilled into super-categories for weak BUs. Handed recommendations to the Business Performance team.
Result	True ROI baseline replacing inflated last-touch numbers; channel healthy overall, so budgets scaled up; SC targeting corrections for weak BUs. Add: % budget scaled, Cr reallocated, ROI range.

Probing questions to rehearse

1. Why 8%? Show the MDE/power logic and the trade-off with revenue lost from the holdout.
2. How did you prevent holdout contamination (users reached on other channels, shared devices)?
3. Incrementality vs attribution: why do they differ and by how much?
4. BU-level results: how did you handle multiple comparisons and small BUs?
5. Why did GMV take longer to reach significance? (variance, heavy tails; mention CUPED or capping)
6. Long-term effects: did you track post-campaign decay or pull-forward?
7. What would you do differently? (pre-registration, geo-level design, always-on holdout, longer horizon)
8. How did you get the business team to act on it? What pushback did you face?

Gaps to fill before the interview: hard numbers for spend, ROI, budget change, and the exact attribution window and metric definitions.

10. Core values & leadership

The pasted description names 'Champion the Mission' and 'Be a Host'. From memory, Airbnb's other core values are 'Embrace the Adventure' and 'Be a Cereal Entrepreneur': verify before the interview.

Theme	Sample question	What they test
Conflict with PM	Disagreed with a PM on what to ship; what happened?	Data-backed position without being combative
Ambiguity	Most ambiguous analysis you solved	Structuring fuzzy asks into a plan quickly
Being wrong	An analysis you got wrong	Honesty, process change
Metric alignment	Two teams used a metric differently	Governance, rallying the org
Prioritization	More requests than capacity	Triage, saying no diplomatically
Building from scratch	Set up an analytics practice	Greenfield ownership
Cross-functional	Fixed a tracking problem with engineers	Operating at the instrumentation layer
Leadership comms	Presented complex analysis to non-technical execs	Compress without losing nuance
Mentoring	Helped an analyst grow; analyst producing insight-poor work	Raising the bar on thinking
Experiment surprise	Test won but you recommended not shipping	Experimentation maturity
Extreme ownership	Time you owned a data outcome end-to-end	Accountability

Story bank (fill with real numbers)

- Meta incrementality: influence, rigor, ownership (also the main presentation).
- Lifestyle-category retention analysis: exploratory driver analysis; frame carefully, outcome was exploratory.
- Churn model with holdout in festive window: ML + experiment, say what you'd improve (uplift).
- A no-ship call on a winning test (build or find one).
- A mentoring story and a PM-conflict story (not yet identified).

Leadership answer template

Situation in one line, your specific action, the pushback you received, how you resolved it, the measurable result, what you learned.

11. Transferable cases from earlier prep

Hike: DAU drop

- Sanity first: YoY/seasonality, pipeline and tracking, releases.
- Segment: new vs existing, platform, geo, channel, cohort age. Example: drop concentrated in 60-90 day users.
- Network-effect hypothesis: contacts and groups; find the 'magic number' elbow by comparing churned vs retained.
- Validate nudges with a randomized holdout; track downstream quality (not just app opens).

Hike: experiments that should not ship

- Do not redefine the north star after results; extending a test post hoc is peeking.
- D30 retention up but sessions down and opt-outs up: the metric was gamed by notifications. Prefer intent-based retention (e.g. sent a message in week 4).
- Fix process: pre-registered metrics, MDE, guardrails, registry.

eBay: pricing & take rate

- Cannot A/B seller prices easily: use geo/category rollouts with DiD; synthetic control for global changes.

- Elasticity: endogeneity, so use historical changes (DiD), thresholds (RD), instruments (IV); segment by seller size.
- Promo ROI: incremental GMV x take rate minus cost; additionality; threshold bunching.
- Uplift vs response: persuadables / sure things / lost causes / sleeping dogs; two-model (T-learner), X-learner, evaluate with Qini/AUUC.

Cross-cutting seniority signals

- Clarify the decision; define success and guardrails before analysis.
- Name second-order effects (supply responds to demand-side changes).
- Never declare success without a valid counterfactual.
- Lead with the so-what: what's happening, why, what if we do nothing, recommendation, how we measure.

12. Mock log & next steps

Item	Status
Round 1 SQL, Q1 occupancy per city	Mock posed; your answer not yet given. Attempt it, then compare with the model answer.
SQL Q2 superhost	Prompt was truncated when pasted; model answer built on the generic schema.
Part A (dataset) A1-A3, Part B B1-B3	Posed with model answers in this PDF.
Travel cases 1-4 + take-home RCA	Frameworks written; not yet mocked.
Take-home dataset analysis	Not run. Approach only.
Questions for your contact	List ready (Section 1).

Suggested next order

1. Answer Q1 under a 14-minute timer, then A1 and A3 (most take-home relevant).
2. Mock the booking-drop case aloud using the Section 3 framework.
3. Practice ITS and synthetic control explanations (5 minutes each).
4. Build the Meta deck with numbers; rehearse the probing questions.
5. Draft 6-8 behavioral stories.